


Procedure #15

Prepared by:
Andy Gomes

Approved by:
Clive Baek

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Testing of the low water cutout devices is required to ensure that they are operating properly. The Cleaver Brooks boiler has three LWCO devices

1. Levelmaster level control is an automatic reset device (LWCO)
2. Drum level sensor probe is a manually reset device (LLWCO)
3. Water Column level control is a manually reset device (LLLWCO)

ASSUMPTIONS:

- The boiler is operating and the water level is in normal operating range.

Step	Operation	Initial when Complete	Date and Time when Complete
1	Open the drain valve on the bottom of level master controller on the Cleaver Brooks boiler slowly while watching the LED lights on the controller start to go out. The levelmaster is located on the left side of the boiler.		
1.1	Ensure that the LWCO alarm comes in after a short delay of 7-10 seconds. After the boiler alarms and shuts off, close the drain valve and restart the boiler.		
2.0	Lower the the drum level in the boiler by opening the bottom blowdown valves slowly and and watch for the level to drop. When the level has dropped enough the red LED light will light up on the LWCO controller which is mounted on the side of the control cabinet. This alarm can also be simulated by pressing the “TEST” button on the LWCO controller which will shutdown the boiler.		

2.1	Ensure that the LWCO alarm comes in and the red LED remains lit. After the boiler alarms and shuts off, close the bottom blowdown valves allow the boiler water to stabilize and reset the button on the controller. Restart the boiler.		
3.0	Open the drain valve on the bottom of the right side water column on the Cleaver Brooks boiler.		
3.1	Ensure that the boiler goes into alarm and shuts down. Close the drain valve and allow the water level to stabilize. Make sure the boiler doesn't restart until the manual reset button has been pressed in the bottom of the water column controller.		
4	Make an appropriate log book entry.		

Note: If the boiler continues to run after any one of these three procedures, please shutdown the boiler and contact the head of mechanical operations.